BUILDING A VR READY PC

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Slide 1:

This report outlines the design and selection of components for a VR-ready PC, built to deliver an immersive, high-performance experience. The system uses top-tier hardware to meet VR demands, ensuring smooth visuals, responsive interactions, and future-proofing for advancing technology.

Slide 2: Agenda Overview

This report covers the essential components for building a VR-ready PC, detailing each part's role in optimizing virtual reality performance. Key topics include the CPU for processing power, GPU for visuals, motherboard for connectivity, and RAM for multitasking. We also examine storage (SSD), power management (PSU), casing, cables, and CPU cooling. Each section highlights how these components contribute to a smooth VR experience.

Slide 3: The Central Processing Unit - Intel Core i9-14900K

The Intel Core i9-14900K is chosen for its high processing power, crucial for managing complex VR tasks [1]. Built on Intel's latest architecture, it achieves impressive clock speeds and efficient multi-threaded performance, ensuring smooth VR experiences [2]. Its advanced thermal management keeps it stable under heavy loads, ideal for gaming and content creation [2]. Benchmark tests confirm its suitability for versatile VR setups, making it a robust choice for demanding applications [3]. This CPU's architecture and performance highlight its effectiveness in high-end gaming and VR environments [1, 2, 3].

Slide 4: The Graphics Processing Unit - NVIDIA GeForce RTX 4090

The NVIDIA GeForce RTX 4090 is the powerhouse GPU chosen for this VR-ready PC, providing unmatched performance for 3D rendering, AI processing, and complex simulations. Equipped with 24GB of GDDR6X memory and built on the advanced Ada Lovelace architecture, this GPU excels in handling high-demand applications efficiently [6]. Its energy-efficient design allows for powerful output with manageable power consumption, making it suitable for both individual setups and data center

environments [4]. With its capability for smooth 4K rendering and high frame rates, the RTX 4090 is ideal for users who need high-definition visuals and robust multitasking in virtual reality [5][6].

Slide 5: The Motherboard - ASUS Maximus Z790 Extreme

The ASUS Maximus Z790 Extreme is a premium motherboard built to support Intel's 14th-gen CPUs, offering advanced features for high-end gaming and productivity. It provides excellent overclocking capabilities, robust power delivery, and efficient thermal management, ideal for maximizing the performance of demanding processors [7][8]. Supporting DDR5 memory, it enables faster data transfers, enhancing system responsiveness for both gaming and intensive applications [9]. With its premium build and extensive connectivity options, this motherboard is a versatile and powerful choice for users who require top-tier performance and future-proofing in their VR setup.

Slide 6: The RAM - Corsair Vengeance RGB 64GB

The Corsair Vengeance RGB 64GB DDR5-6000 is chosen for its high speed and low latency, providing a performance boost with faster load times and smoother gameplay in VR environments [22]. This RAM is optimized for both AMD and Intel platforms, supporting easy overclocking with AMD EXPO and Intel XMP technologies, making it versatile across setups [23]. Additionally, its customizable RGB lighting and high-quality heat spreaders add style and stability, enhancing the system's aesthetics and performance [24].

Slide 7: The SSD - Samsung 980 PRO

The Samsung 980 PRO SSD offers unmatched speed in the NVMe market, with read speeds up to 7,000 MB/s and write speeds up to 5,000 MB/s, making it ideal for VR applications with heavy data loads [18]. Its advanced heat management system, including Dynamic Thermal Guard and optional heat sinks, ensures stable performance during extended sessions [16]. Additionally, Samsung's V-NAND technology and Power Loss Protection enhance reliability and data integrity, ensuring long-term durability [17].

Slide 8: The Power Supply Unit - Corsair AX1200i

The Corsair AX1200i power supply unit delivers over 92% efficiency at 50% load due to its 80 PLUS Platinum certification, which minimizes power waste and heat output, essential for high-performance builds [20][21]. Its build quality features durable Japanese capacitors, ensuring reliable performance even under heavy loads [19]. Additionally, the AX1200i's zero RPM fan mode keeps the unit silent under low to moderate loads, making it ideal for quiet, high-performance systems [21].

Slide 9: The Casing - Meshify 2 XL ATX Full Tower Case

The Meshify 2 XL ATX case from Fractal Design is renowned for its outstanding airflow, featuring a mesh front panel that maximizes intake and exhaust to keep components cool under heavy loads [10]. It

provides ample space for versatile expansion, allowing the installation of large GPUs, multiple radiators, and various storage drives, making it ideal for high-performance setups [11]. Additionally, its design supports efficient cable management with space behind the motherboard tray and built-in routing options, enhancing organization and airflow [12].

Slide 10: The CPU Cooler - Noctua NH-D15 chromax.black

The Noctua NH-D15 chromax.black provides exceptional cooling performance with its dual tower design and six heat pipes, making it ideal for overclocked CPUs in high-performance systems [13]. Its ultra-quiet operation, featuring NF-A15 PWM fans, ensures minimal noise, which is advantageous for immersive VR experiences [15]. Known for premium build quality, Noctua products like the NH-D15 are durable, offering long-lasting reliability [14].

Slide 11: Additional Components

The Noctua A14 PWM chromax.black.swap fan provides high airflow (82.52 CFM) with PWM control, allowing balanced cooling and noise management. Its durable design and anti-vibration pads enhance airflow and stability in the VR setup. The Noctua NT-H1 thermal paste ensures efficient heat transfer between the CPU and cooler, featuring easy application and long-lasting stability, which is essential for maintaining optimal CPU temperatures in demanding VR applications.

Conclusion:

Our VR-ready PC is build optimized for performance and immersion, featuring key components like the Intel Core i9-14900K, NVIDIA RTX 4090, ASUS Maximus Z790 motherboard, and Corsair Vengeance RAM, provides powerful processing, graphics, and multitasking capabilities. Reliable power from the Corsair AX1200i and efficient cooling solutions ensure stability, making this setup ideal for immersive VR experiences.

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Presentation Canva Link:
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